

Reese Ludwick

reesepludwick@gmail.com | (304)-886-1850 | Remote LinkedIn: <https://www.linkedin.com/in/reese-ludwick-024333286/> | GitHub: <https://github.com/reese8272>

Summary

Applied AI / backend engineer who turns models into reliable production behavior: agent orchestration, async pipelines, RAG, and MCP integrations with attention to latency, cost, and correctness. Ships GenAI/agent systems at Cognizant (LangChain / LangGraph) plus AI-augmented full-stack for clients — caching cuts token cost, async workers keep latency low.

Projects

CreatorClip / autoclip.studio

Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg, YouTube OAuth - Live SaaS (Stripe billing); async Celery runs transcription, embedding, rendering off the request path for a responsive API. - WhisperX transcription; Voyage AI embeddings in pgvector for semantic retrieval; ffmpeg media processing. - Per-creator Postgres RLS isolates tenants; creator “DNA” tunes output to channel style; YouTube OAuth. - Cost/latency-aware: heavy model and media steps are decoupled, queued, and independently scalable.

CFO Agent — *in active development*

Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API - FastAPI + LangGraph agent, PostgreSQL persistence, Redis caching; CSV/OFX import + encrypted vault for financial data. - Hybrid context architecture; multi-tenant-ready design. - Redis caching cuts repeated-call latency and token/compute cost on the agent’s hot paths.

Agentic Software Development Team

LangChain, LangGraph, Amazon Bedrock — *Cognizant R&D* - Multi-agent LangGraph system with specialized codegen, testing, security, and supervision agents. - Amazon Bedrock plus open-source ML backends for model flexibility.

Customer Call Data Analysis & Insights Engine

Python, NLP, Sentence Transformers — *Cognizant R&D* - NLP clustering of call data into semantic issue groups using Sentence Transformers. - Agentic AI layer for call consolidation and executive-facing reporting.

Experience

Software Associate — Cognizant (Remote)

Dec 2025 – Present - Build GenAI and agentic AI systems with LangChain, LangGraph, and Python across R&D and client deliverables. - Develop RAG pipelines and MCP server integrations for production workflows. - Scope, prototype, and deliver AI data tools cross-functionally.

Freelance AI Engineer & Full-Stack Developer — WorldCovers / APMC (Remote)

May 2026 – Present - Ship production web properties and the WorldCovers platform: features/automation on a 14-table PostgreSQL schema. - Django REST + React/TypeScript; AI-augmented engineering (Claude Code) on tight timelines with stakeholder coordination.

GenAI Developer Contractor — Cognizant (Remote)

Jun 2025 – Dec 2025 - Built GenAI POCs with LangChain, LangGraph, and Python; prompt engineering and model evaluation. - Built custom FastMCP servers and integrated external MCP servers with AWS for dynamic agent tool access. - Delivered modular, production-style agentic workflows.

IT Student Helper — James Rumsey Tech

2019 – 2021 - Resolved 300+ support tickets; led a Wi-Fi overhaul that cut latency 25%; completed 100+ AV/network setups.

Education

BS Computer Science — **Shepherd University** (Aug 2021 – May 2025) Major GPA 3.43

Technical Skills

Languages: Python, SQL **AI / Agentic:** LangChain, LangGraph, RAG, MCP Servers, Amazon Bedrock, Sentence Transformers, TensorFlow, pgvector, Anthropic API **Backend / Infra:** FastAPI, Celery, Redis, PostgreSQL, Docker, Django REST **Data:** Pandas, NumPy, Jupyter **Frontend:** React / TypeScript **Tooling / Platform:** Linux, Ubuntu, Git, GitHub, Azure Fundamentals, VS Code, Claude Code

Certifications

- TensorFlow Developer Specialization
- Linux + Git (Boot.dev)
- AZ-900 (Azure Fundamentals)
- GitHub Copilot
- Generative AI Context Engineering (Cognizant)